

“Greenhouse Monitoring System Using ESP8266”

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**GOVERNMENT COLLEGE OF ENGINEERING NAGPUR
DEPARTMENT OF
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CERTIFICATE

This is to certify that the Mini Project entitled “**Greenhouse Monitoring System using ESP 8266**” which is being submitted here with for the award of Third Year B. Tech, is the result of the work completed by
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This is our privilege to express our deep sense of gratitude and indebtedness to our project

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ABSTRACT

The Greenhouse Monitoring System is a mini project aimed at developing a low-cost and efficient solution to monitor and control environmental parameters within a greenhouse environment. This system utilizes a DHT11 sensor for temperature and humidity measurement, and a gas sensor (MQ series) to detect harmful gases such as ammonia or methane, which can negatively impact plant health [1][2]. These sensors are interfaced with the ESP8266 microcontroller, which processes the sensor data and displays it on an LCD screen or transmits it wirelessly to a cloud platform for real-time monitoring [3]. The ESP8266 also controls actuators such as exhaust fans or alarms based on preset threshold values. This setup enables remote access and monitoring of greenhouse conditions, reducing the need for constant manual oversight. The system highlights the potential of IoT-based solutions in modern agriculture by providing automated, real-time environmental control to enhance plant growth and sustainability.

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CHAPTER 1

INTRODUCTION

Greenhouses have revolutionized modern agriculture by providing a controlled environment for plant growth, ensuring better yield, quality, and protection from external factors such as extreme weather and pests. However, maintaining an optimal greenhouse environment requires continuous monitoring and control of crucial parameters like temperature, humidity, and gas concentration. Any deviation from the ideal conditions can affect plant health, leading to reduced productivity and financial losses for farmers. Additionally, fire hazards due to electrical malfunctions or overheating pose significant risks to greenhouse infrastructure and crops.

To address these challenges, our project, "Greenhouse Monitoring System using ESP8266," aims to provide an efficient and automated solution for real-time monitoring of environmental conditions inside a greenhouse. The system is built using the ESP8266 microcontroller, which enables wireless connectivity and remote monitoring. It incorporates multiple sensors, including the DHT11 for temperature and humidity measurement and a gas sensor for detecting harmful gases. The collected data is analysed to ensure optimal growing conditions, and in case of any abnormal changes, the system triggers alerts for immediate action. The primary objective of this project is to enhance greenhouse management by reducing manual intervention and providing an automated system that ensures the necessary environmental conditions are maintained. Traditional greenhouse monitoring methods require frequent physical checks, which can be time-consuming and inefficient. By integrating IoT-based technology with environmental sensing, our system allows for remote monitoring and timely alerts, thus improving overall efficiency and productivity.

One of the major concerns in greenhouse operations is fire hazards, which can result from overheating or electrical faults. Our system incorporates fire detection capabilities using temperature and gas sensors. If the temperature exceeds a critical threshold or if a dangerous gas is detected, the system activates a buzzer and LED alerts to notify users immediately. This feature significantly enhances safety by allowing quick responses to prevent potential disasters. The need for such a system is particularly relevant in the context of climate change and the growing demand for smart agricultural solutions. As environmental conditions become increasingly unpredictable, ensuring the proper management of greenhouse parameters is more critical than ever. This project supports sustainable agriculture by optimizing resource usage, reducing energy wastage, and minimizing crop loss due to unfavourable environmental factors.

CHAPTER 2

AIM & OBJECTIVES

2.1 Aim

To design and implement an IoT-based Greenhouse Monitoring System using ESP8266 to continuously monitor temperature, humidity, and gas levels while providing fire alerts for enhanced safety and optimal plant growth.

2.2 Objectives

1. To monitor environmental parameters – Measure temperature, humidity, and gas levels inside the greenhouse in real time using appropriate sensors.
2. To provide real-time alerts – Notify users immediately through buzzer and LED indicators in case of abnormal conditions or fire hazards.
3. To enhance greenhouse safety – Detect potential fire hazards by monitoring temperature and gas concentration, ensuring timely intervention.
4. To reduce manual supervision – Automate the monitoring process, minimizing the need for frequent manual checks and improving efficiency.

The primary objective of this project is to develop an IoT-based Greenhouse Monitoring System using ESP8266 that ensures continuous monitoring of environmental parameters such as temperature, humidity, and gas levels. By integrating sensors with a microcontroller, the system provides real-time data and alerts to maintain optimal greenhouse conditions. One of the key goals is to enhance greenhouse safety by detecting fire hazards through temperature and gas concentration monitoring, triggering immediate alerts via buzzer and LED indicators. Additionally, the project aims to reduce the need for manual supervision by automating the monitoring process, allowing farmers and greenhouse operators to access data remotely through IoT technology. This automation not only improves efficiency but also contributes to sustainable agriculture by preventing crop losses, optimizing resource usage, and ensuring a stable growing environment. Ultimately, the system provides a smart and cost-effective solution for modern greenhouse management.

CHAPTER 3

LITERATURE SURVEY

i-learning IoT: An intelligent self-learning system for home automation using IoT, Vishwajeet Hari Bhide, Sanjeev Wagh [4] proposed an efficient implementation for IoT (Internet of Things) used for monitoring and controlling the home appliances via World Wide Web. Automation of regular activities inside the home is home automation. Due to huge advancement in wireless sensor network and other computation technologies now a day's, it is possible to provide flexible and low cost home automation system. In this system prediction is done to find out the required solution if any problem occurs in any device connected to the system. The home appliances can be controlled via Smartphone using Wi-Fi. Here ESP8266 used as server system and Wi-Fi as communication protocol.

IoT based monitoring and control system for home automation, D. Pavithra, Ranjith Balakrishnan [5] states features of their system. An extra feature that enhances the facet of protection from fireplace accidents is its capability of sleuthing the smoke in order that within the event of any fireplace, associates an alerting message and an image is sent to Smartphone on Blynk App. The design and implementation of the greenhouse monitoring system based on GSM and RF technologies, Zhao, Zheng Xiangyang, Duan Chen, Chen Zhaohui Sang Shangming, Zhang Zhaohui [4] implemented small and medium size greenhouse environmental monitoring. Using ATmega16A, low-power chip PTR4000 and Huawei wireless module GTM900-C to design monitoring nodes and sink nodes of the system. Advantages of this system are stable, reliable, and it is able to achieve real-time monitoring of greenhouse environment.

Design of intelligent greenhouse environment monitoring system based on ZigBee and embedded technology, WeiminQiu, Linxi Dong, Fei Wang, Haixia Yan [1] designed intelligent greenhouse environment monitoring control system which is based on ZigBee and embedded technology. This system consists of three main components: upper machine processors, environmental factors acquisition nodes and intelligent control terminal block. It is mainly made up of S3C2440 microprocessor with the ARM920T core, power, clock, reset circuit, LCD touch screen, intelligent control terminal module, ZigBee coordinator and environmental factors detection terminal nodes.

CHAPTER 4

METHODOLOGY

4.1 Block Diagram

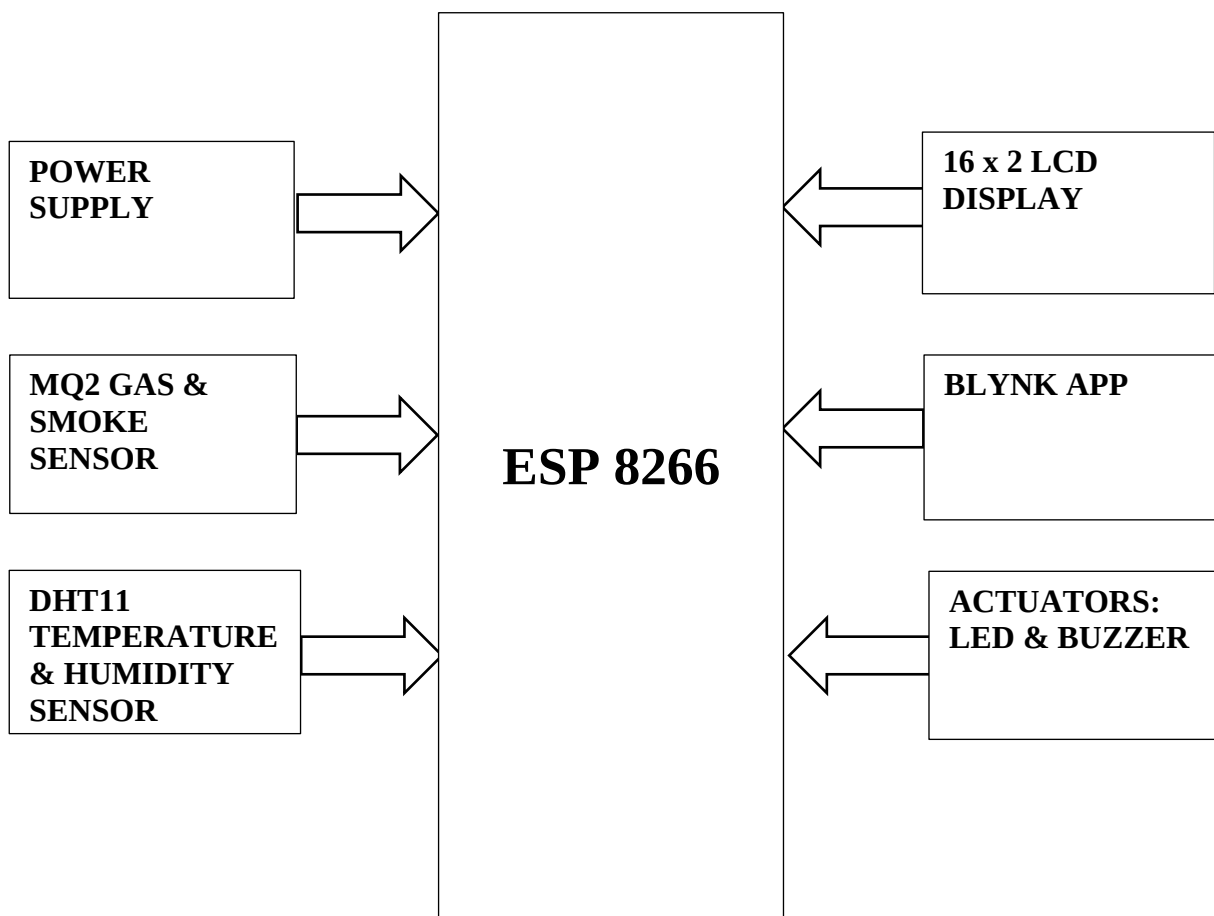


Fig. 4.1: Block Diagram of Greenhouse Monitoring System using ESP 8266

4.2 Hardware Description

Sr.	Components	Specifications	Quantity
1.	ESP 8266	Processor :-32bit Tensilica L106 RISC Clock Speed :-80 MHz RAM :- 80 KB user + 36 KB data	1
2.	DHT11 Temperature & Humidity Sensor	Operating voltage :- 3.3V - 5V Pin count :- 4 pins	1
3.	MQ2 Gas Sensor	Detection range :- 20 – 10,000 ppm Temperature range :- -10°C to 50°C Outputs :- Analog & Digital output	1
4.	Buzzer	Sound output :- 85 – 110 dB	1
5.	LED	3V White LED	1
6.	16x2 LCD Display	Operating voltage ~5V	1
7.	Blynk App	Cloud based Mobile platform	-
8.	Power Supply	9V DC for ESP 8266	1
9.	Connecting Wires	-----	

Table 4.2: Hardware Components Used in Greenhouse Monitoring System using ESP 8266

4.2.1 NodeMCU ESP 8266 Microcontroller

The ESP8266 block diagram consists of two main sections: the Analog Wi-Fi Block and the Digital Block. The Analog Wi-Fi Block manages wireless communication, including RF transmission and reception, power control, and frequency generation via a PLL and VCO. It integrates an RF balun and antenna switch for efficient signal processing. The Digital Block houses the Tensilica L106 32-bit CPU, SRAM for memory, and various I/O interfaces such as ADC, GPIO, I²C, PWM, SPI, and UART for connecting sensors and actuators. Additionally, the digital baseband converts signals for secure Wi-Fi communication. Together, these blocks enable efficient data processing and wireless connectivity, making ESP8266 ideal for IoT applications like your Greenhouse Monitoring System.

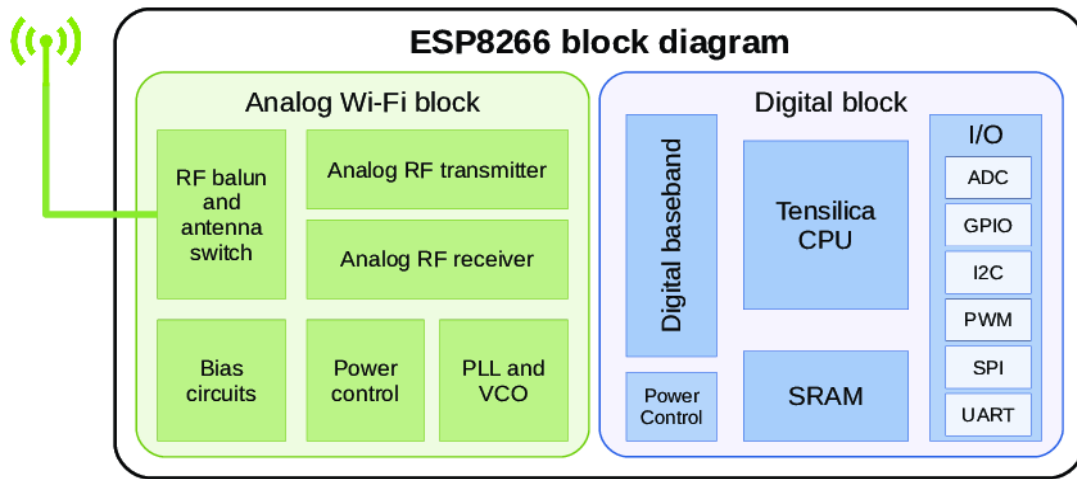


Fig. 4.2.1: Block Diagram of ESP 8266 Microcontroller

4.2.2 DHT11 Temperature and Humidity Sensor

DHT11 sensor consists of a capacitive humidity sensing element and a thermistor for sensing temperature. The humidity sensing capacitor has two electrodes with a moisture holding substrate as a dielectric between them. Change in the capacitance value occurs with the change in humidity levels. The IC measure, process this changed resistance values and change them into digital form. For measuring temperature this sensor uses a Negative Temperature coefficient thermistor, which causes a decrease in its resistance value with increase in temperature. To get larger resistance value even for the smallest change in temperature, this sensor is usually made up of semiconductor ceramics or polymers.

The temperature range of DHT11 is from 0 to 50 degree Celsius with a 2-degree accuracy. Humidity range of this sensor is from 20 to 80% with 5% accuracy. The sampling rate of this sensor is 1Hz.i.e. it gives one reading for every second. DHT11 is small in size with operating voltage from 3 to 5 volts.

The maximum current used while measuring is 2.5mA.

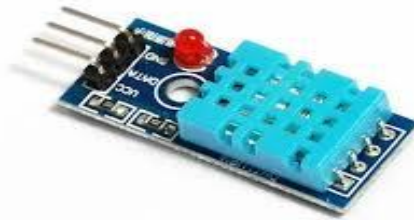


Fig. 4.2.2: DHT11 Temperature and Humidity Sensor

4.2.3 MQ2 Gas Sensor

The MQ2 is a heater-driven sensor. It is therefore covered with two layers of fine stainless steel mesh known as an “anti-explosion network”. It ensures that the heater element inside the sensor does not cause an explosion because we are sensing flammable gasses. It also protects the sensor and filters out suspended particles, allowing only gaseous elements to pass through the chamber. A copper-plated clamping ring secures the mesh to the rest of the body. To summarize, the Heating System is composed of a Nickel-Chromium coil and an Aluminum Oxide-based ceramic, while the Sensing System is composed of Platinum wires and a Tin Dioxide coating.



Fig. 4.2.3: MQ2 Gas sensor

4.2.3 16x2 LCD Display

This compact display, featuring a resolution of 16 characters across 2 lines, offers a cost-effective and user-friendly method for visual interaction with projects. Its primary function lies in presenting messages, sensor readings, or user interface elements clearly. The popularity of 16x2 LCDs stems from their ease of use. They typically communicate with microcontrollers using a minimal number of pins, making them ideal for beginners to learn embedded systems programming fundamentals and user interface design principles. Furthermore, the widespread adoption of 16x2 LCDs translates to a wealth of readily available online resources and tutorials, streamlining the integration of visual feedback into embedded system projects. This ease of use and extensive support make 16x2 LCDs a valuable tool for both learning and development in the embedded systems domain.



Fig. 4.2.4 16X2 Lcd Display

4.3 Software Description

4.3.1: Arduino Integrated Development Environment (IDE)

The Arduino IDE (Integrated Development Environment) is a crucial tool for developing and uploading code to the ESP8266 in your Greenhouse Monitoring System project. It provides a simple interface to write, compile, and upload C/C++ code to microcontrollers like the ESP8266. Here, the Arduino IDE helps in interfacing with the DHT11 temperature sensor, gas sensor, LDR sensor module, buzzer, and LEDs by using appropriate libraries such as DHT.h for temperature and humidity readings. The ESP8266WiFi library can be used to connect the system to a network if remote monitoring is needed. Additionally, the Serial Monitor in Arduino IDE is useful for debugging sensor data and ensuring that the system responds correctly to environmental changes, such as activating alerts for high

gas levels or fire hazards.

To use the ESP8266 with Arduino IDE, some initial setup is required. After installation, selecting the correct ESP8266 board model and COM port is necessary to ensure proper communication between the microcontroller and the computer. Once the setup is complete, coding begins by defining pin configurations and initializing the required sensors. Uploading the code to the ESP8266 is a straightforward process using the Upload button in Arduino IDE. Once uploaded, the system continuously collects and processes sensor data, responding to changes in the environment.

4.3.2: Blynk App

Blynk is a powerful IoT platform that enables remote monitoring and control of microcontroller-based projects, making it an ideal choice for the Greenhouse Monitoring System. It allows users to visualize real-time sensor data, such as temperature, humidity, and gas levels, directly on a smartphone. With its user-friendly interface, Blynk provides widgets like gauges, graphs, and notifications, helping users track environmental conditions efficiently. The ESP8266 connects to Blynk via WiFi, sending sensor readings to the app, which can also trigger alerts if values exceed predefined thresholds.

In this project, Blynk enhances automation and remote accessibility, reducing the need for constant manual monitoring. Users can configure alerts to receive notifications when gas levels are too high or when fire hazards are detected. Additionally, actuators like LEDs and buzzers can be controlled remotely via the app. The Blynk library in Arduino IDE facilitates seamless communication between the ESP8266 and Blynk Cloud, ensuring reliable data transfer. By integrating Blynk, the Greenhouse Monitoring System becomes a smart, real-time, and user-friendly solution for maintaining an optimal greenhouse environment.

4.3.3: C++ Programming Language

The Greenhouse Monitoring System is programmed using C++, a widely used language in embedded systems and IoT applications. C++ provides a structured and efficient way to interact with hardware components like sensors, actuators, and communication modules. In this project, C++ is used to read data from the DHT11 temperature sensor, gas sensor, and LDR module, process the values, and trigger actions like activating LEDs or a buzzer when critical thresholds are reached. The language's ability to handle real-time sensor input, conditional statements, and loops makes it ideal for implementing automated responses in the system.

By leveraging C++ in Arduino IDE, the project benefits from predefined libraries that simplify complex tasks. For example, the DHT library makes it easy to extract temperature and humidity data, while the Blynk library facilitates WiFi communication for remote monitoring. Object-oriented features of C++ allow for better code organization and modularity, making debugging and future enhancements easier. With C++ handling the core logic, the ESP8266 efficiently processes sensor data.

4.4 Working

1. Sensor data collection

The DHT11 sensor continuously measures temperature and humidity in the greenhouse. The gas sensor detects the presence of harmful gases and triggers an alert if the concentration exceeds a set limit.

2. Data Processing and Decision Making

The ESP8266 microcontroller, programmed using C++ in Arduino IDE, collects real-time data from all sensors. Based on predefined threshold values, the system decides whether to activate alerts. For example, if the gas sensor detects a hazardous level of gas, the buzzer and LED turn on, warning users of a potential danger.

3. Remote Monitoring via Blynk App

The ESP8266 connects to the Blynk Cloud via WiFi and sends real-time sensor readings to the Blynk app on a smartphone. Users can view live data on temperature, humidity, gas levels, and light intensity through graphical widgets like gauges and graphs. If any value exceeds the safe limit, the app sends a notification alert to the user, allowing them to take necessary action.

4. Continuous Monitoring and Data Logging

The system continuously operates, ensuring real-time monitoring of greenhouse conditions. Data can be logged on Blynk Cloud, providing historical records of environmental changes. This allows users to analyse trends and optimize greenhouse conditions for better plant growth.

Circuit Diagram:

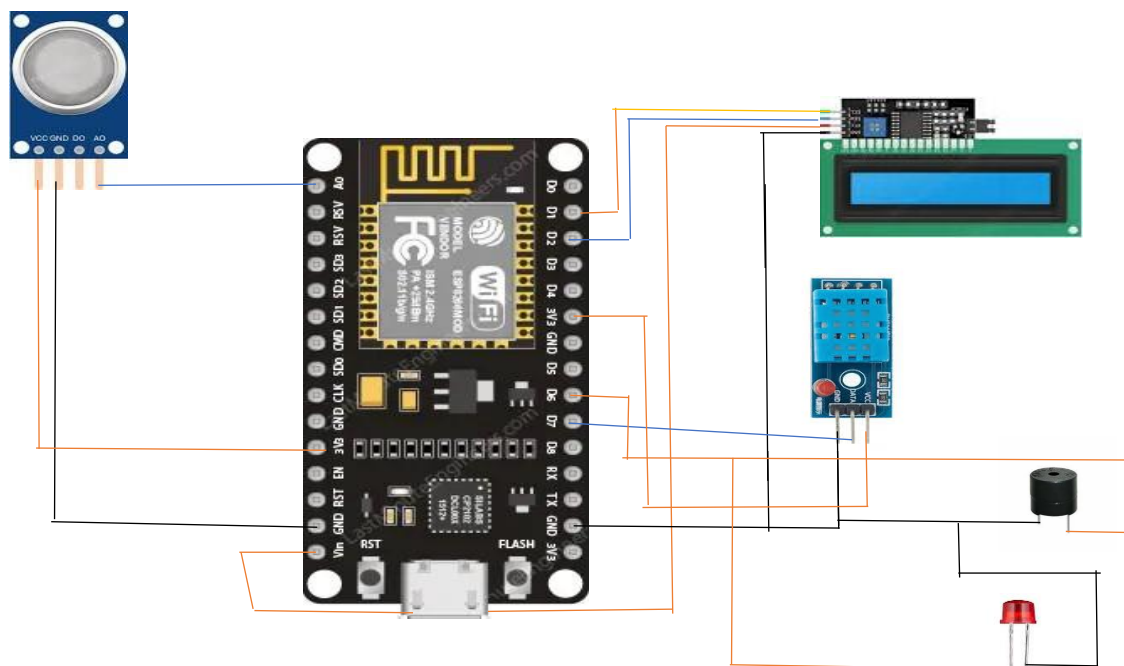


Fig 4.4.1 Circuit Diagram of Greenhouse Monitoring System using ESP826

CHAPTER 5

RESULTS & DISCUSSIONS

5.1 Results

The Greenhouse Monitoring System using ESP8266 successfully monitors and displays real-time temperature, humidity, gas levels, and light intensity. By integrating DHT11, a gas sensor, and an LDR module, the system continuously tracks environmental conditions and provides instant feedback. The ESP8266 microcontroller processes sensor data and effectively communicates with the Blynk app, allowing users to access live data remotely on their smartphones. This ensures better greenhouse management with minimal manual intervention. The system's ability to provide real-time monitoring, instant alerts, and historical data analysis makes it a reliable and efficient solution for automated greenhouse management. Future improvements, such as automated irrigation and AI-based analytics, could further enhance its capabilities, making greenhouse farming smarter and efficient.

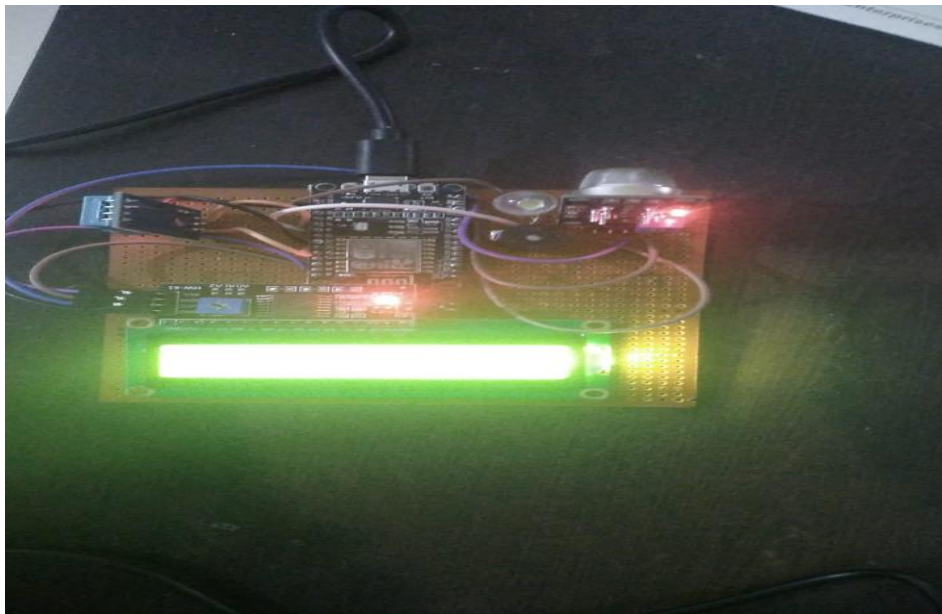


Fig. 5.1.1: Greenhouse Monitoring System using ESP 8266

Once the Greenhouse Monitoring System is powered on, the 16x2 LCD displays real-time monitoring results, showing the temperature, humidity, and light levels measured by the sensors. This display provides immediate feedback on the environmental conditions within the greenhouse, making it easy for users to see the data without needing to access the Blynk app. The LCD continuously updates to reflect the current values, ensuring that the system's status is always visible. If any of the monitored parameters exceed the set thresholds, the system triggers alerts using LEDs and a buzzer, but the LCD also shows a warning message to provide additional clarity. This local display feature enhances the system's functionality, offering both remote monitoring through the Blynk app and a direct, on-site visual representation of greenhouse conditions.

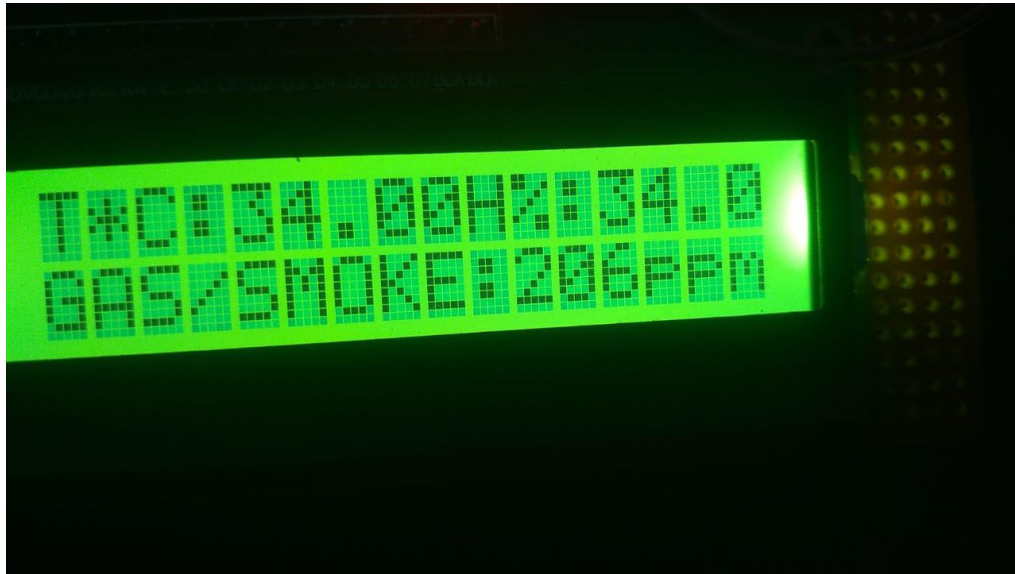


Fig.5.1.2: LCD Displaying Gas, Temperature and Humidity levels (in ppm)

When gas or smoke is detected near the gas sensor, the system responds immediately by activating the buzzer and causing the LED to glow. This serves as a local alert mechanism, warning anyone nearby of the potential hazard. The buzzer sound is loud enough to catch attention, while the LED not only glows but also displays a fire message to indicate a serious threat, ensuring users are aware of the situation at a glance. This provides a quick, visual, and audible warning that can prevent further escalation of the danger. In addition to the local alerts, the system sends a notification to the Blynk app, allowing users to receive real-time updates even if they are not physically present at the greenhouse. The notification alerts the user to the elevated gas or smoke levels, ensuring they can take immediate action remotely, such as activating ventilation or contacting emergency services. This combination of local and remote alerts makes the Greenhouse Monitoring System highly effective in preventing fire hazards and ensuring the safety of the greenhouse environment.

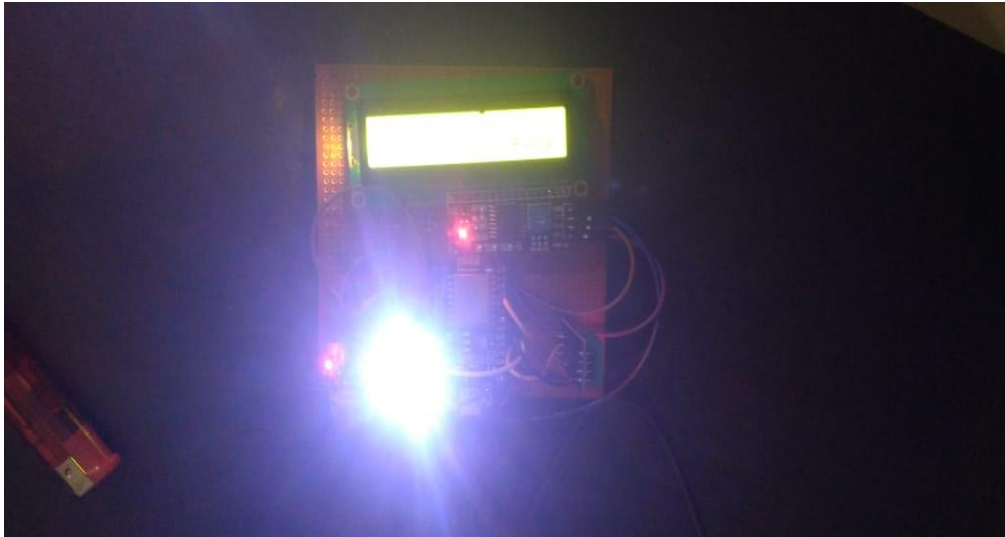


Fig. 5.1.3: Working of LED and Smoke sensor after it detects smoke

The Blynk IoT app plays a crucial role in the Greenhouse Monitoring System by displaying real-time data from the sensors. Once the system is connected to WiFi through the ESP8266, the app provides an intuitive interface where users can monitor key environmental parameters such as temperature, humidity, gas levels, and light intensity. The app visualizes the data through gauges, graphs, and other widgets, providing a clear and interactive representation of the greenhouse conditions. Users can monitor these parameters remotely, ensuring the greenhouse environment is optimal for plant growth without needing to be physically present. In addition to real-time monitoring, the Blynk app is responsible for sending immediate notifications when any of the monitored parameters exceed predefined thresholds. If the system detects high gas levels or smoke, for example, the app alerts the user via a notification so they can take quick action. The integration with Blynk IoT enhances the system's accessibility and control, making it easier for users to manage their greenhouse remotely, from anywhere in the world. This real-time display of data and alerts ensures that the greenhouse environment is constantly monitored, improving both safety and efficiency.

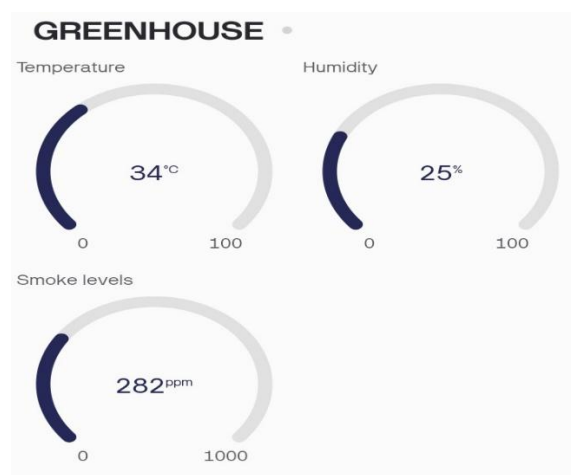


Fig. 5.1.4 Blynk Monitoring Display

5.2 Discussions

The Greenhouse Monitoring System demonstrates the power of IoT technology in modernizing agriculture and improving the efficiency and safety of greenhouse management.

1. **Real-Time Monitoring:** The system provides continuous monitoring of key parameters such as temperature, humidity, gas levels, and light intensity. These readings are sent in real-time to the Blynk IoT app, allowing users to remotely track greenhouse conditions at all times.
2. **Alert Mechanism:** The gas sensor detects hazardous gas levels or smoke, triggering a buzzer sound and an LED glow with a fire message. This provides a local alert mechanism to warn users immediately of potential dangers like gas leaks or fire hazards. The system also sends a notification to the Blynk app, ensuring users are alerted even when they are not on-site.
3. **WiFi Communication and Reliability:** The ESP8266 microcontroller ensures real-time communication between sensors and the Blynk app via WiFi. However, the effectiveness of the system depends on the stability of the WiFi connection. Poor or intermittent WiFi signals can cause delays in data transmission or missed notifications, affecting the system's performance.
4. **Scalability and Flexibility:** The system is scalable, allowing for future upgrades. Adding more sensors and control systems can extend the system's capabilities, turning it into a more sophisticated solution for automated greenhouse management.

The project enhances greenhouse safety, efficiency, and automation, reducing risks from fire and harmful gases while maintaining optimal growing conditions for plants.

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 Conclusion

The Greenhouse Monitoring System using ESP8266 successfully provides an efficient and cost-effective solution for monitoring and maintaining optimal greenhouse conditions. By integrating sensors such as DHT11 (temperature & humidity), gas sensor, and LDR, the system ensures real-time environmental tracking. The buzzer and LED alerts, along with Blynk app notifications, enhance safety by quickly notifying users of hazardous conditions like gas leaks or fire.

This project demonstrates the potential of IoT-based smart agriculture, helping farmers and researchers maintain a controlled environment with minimal manual intervention. With further enhancements, such as automatic ventilation and irrigation control, this system can evolve into a fully automated smart greenhouse solution.

6.2 Future Scope

1. **Integration with Automated Control Systems:** Implement automatic ventilation or irrigation systems based on temperature and humidity levels and use motorized windows or fans to regulate greenhouse climate automatically.
2. **AI and Data Analytics:** Implement machine learning models to predict environmental changes and optimize greenhouse conditions. Also store historical data for trend analysis and better decision-making.
3. **Solar Power Integration:** Use solar panels to make the system energy-efficient and sustainable and implement a battery backup system for uninterrupted operation.
4. **Mobile App & Web Dashboard Enhancement:** Develop a dedicated mobile app with real-time graphs, alerts, and recommendations. Implement a cloud-based system for remote monitoring from anywhere in the world.
5. **IoT and Smart Agriculture Expansion:** Integrate with other IoT-based agricultural systems for large-scale monitoring. Enable voice assistant support (Alexa, Google Assistant) for easy control.

With these improvements, the Greenhouse Monitoring system can evolve into a fully

automated, AI-driven smart greenhouse system, helping farmers achieve higher crop yields, lower costs, and better resource management.

CHAPTER 7

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